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Lab 1

Event Analysis

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Introduction

The purpose of this lab is to investigate a network capture file with potentially suspicious activity. Using digital forensics tools such as Wireshark, NetworkMiner, and Snort we are able to complete a deep analysis of the network activity. These tools provide us with the ability to examine traffic flows, reconstruct files, extract credentials, and analyze alerts. This process provides practical experience handling cybersecurity incidents by simulating the workflow of a real world scenario.

Methodology

Wireshark Analysis

After locating the pcap file, I began analyzing it with wireshark. My first step was to go to Statistics and then Capture File Properties.

Here I was able to identify the duration, packet count, and total bytes. The session took a total of 8 minutes and 25 seconds, captured a total of 2,449 packets, and captured 811,157 bytes. For the correlating image see [Figure 1.1](#) in the Wireshark Analysis Appendix.

My next step was to go to Statistics and then Protocol Hierarchy to discover what protocols were present.

The capture contained several common protocols. At the link layer, all traffic was carried over Ethernet. At the network layer, IPv4 was observed, with a small percentage of ARP traffic. At the transport layer, both TCP and UDP were used, with TCP accounting for the majority of traffic (84.9%). Within TCP, protocols such as TLS, HTTP, and FTP were present. UDP traffic included DNS and NetBIOS name services, as well as TiVoConnect discovery traffic. For an image of the referenced data, see [Figure 1.2](#) in the Wireshark Analysis Appendix.

My next step was to identify when the bulk of the data was transferred, so I went into Statistics and then into the I/O Graphs.

Analysis of the I/O Graph shows that the majority of data transfer occurred in bursts rather than being evenly distributed across the 8-minute capture. The largest packet spike occurred between 17:31 and 17:32, when traffic peaked at over 150 packets per second, indicating a period of heavy network activity. The I/O graph also displayed several TCP analysis flags (shown as red bars), primarily clustered at 17:31:05 to 17:31:20, around the same time as the major packet spikes. These errors likely indicate packet retransmissions or out-of-order segments, suggesting that the network was

experiencing stress during peak activity. See [Figure 1.3](#) in the Wireshark Analysis Appendix for packets over time.

To better analyze when the bulk of data was transferred, I set the Y-axis to bytes/sec instead of packets/sec. This revealed several distinct bursts of heavy traffic. The first large data transfer correlates with the heavy packet transfer between 17:31 and 17:32, capping at about 50,000 to 60,000 bytes/sec. However, it also revealed a second larger spike around 17:37:30 where it reached over 90,000 bytes/sec. See [Figure 1.4](#) in the Wireshark Analysis Appendix for bytes per second.

My next step was to figure out what caused the spikes. To do so, I applied a filter (`frame.time >= "Oct 30, 2005 17:31:00" && frame.time <= "Oct 30, 2005 17:31:20"`) to get a look at the packets in the spike. See [Figure 1.5](#) in the Wireshark Analysis Appendix.

I then opened conversations with the filter applied. Analysis of the TCP Conversations revealed that nearly all traffic originated from a single client (172.16.1.35) communicating with an external web server (216.166.24.20). The largest spikes corresponded to multiple concurrent HTTP sessions on port 80, with individual streams transferring 20–35 KB each in less than half a second. The traffic direction was predominantly server-to-client, confirming that the spikes were caused by the client downloading web objects.

This is consistent with the Protocol Hierarchy results over the filtered time, where HTTP accounted for 42.7% of bytes captured, making it the primary driver of network bursts. TCP retransmissions observed in the I/O Graph aligned with these downloads, suggesting the network was briefly congested during peak transfers.

Next I investigated the spike in bytes around 17:37:30. Following the same method, I applied a filter and looked at the conversation. Following the TCP stream with the most bytes revealed interactions with American Online (AOL). It appears to be an installation/update of AOL software. However, suspicious activity did take place. Multiple EXEs and DLLs were downloaded directly via HTTP. There were also registry keys/system DLL references confirming that system-level modifications were being made. In addition, the inclusion of advertising/search traffic (Amazon, AOL shopping, AOL radio, etc.) indicates this was not just a core AOL client, but possible bloatware/adware-type modules being included with the AOL installation.

Further analysis revealed that the host ip address (172.16.1.35) is associated with the device named “KAUFMANUPSTAIRS.” The device is running Windows XP (pc/WinNT:5.1.2600) as its operating system. The device is also broadcasting TiVoConnect Discovery Protocol over UDP 2190, advertising a TiVo Media Server on port 8080 (HTTP), and revealing its UUID ({9625E281-0AD4-4D95-8735-F59AB074E79A}).

NetworkMiner Analysis

Analyzing the file with Network Miner revealed all of the hosts present in the captured traffic. Most traffic looks normal (AOL, MSN, Yahoo) for 2005 home usage. However, rfbcu.org (banking) and Linux-wlan.org are worth extra attention. They could be indicators of sensitive data exposure or potential malicious activity if unexpected.

To check what data may have been leaked, I took a look at the credentials tab. It revealed that the host, 172.16.1.35 (KaufmanUpstairs), was responsible for all the logins to the following servers:

- Microsoft (207.46.19.60, home.microsoft.com)
- Linux-wlan.org (FTP, anonymous login)
- MSN (65.54.104.158, 209.3.40.190)
- UTSA faculty server (129.115.21.158)
- Yahoo mail (216.218.71.184, 206.190.37.187, 206.190.37.187)
- Yahoo bc.us.yahoo.com

See [Figure 2.1](#) for a screenshot of the servers accessed by the host device.

The one that stands out the most is: Linux-wlan.org (FTP, anonymous login). It is associated with the external host address 66.39.92.157. The Anonymous FTP login has the username: anonymous and password: IEUser@. This is a classic anonymous FTP (File Transfer Protocol) login, suggesting there may have been something downloaded or uploaded.

The next step was to analyze the file tab. In addition to normal web content (GIFs, CSS, and JavaScript), suspiciously named HTML files (redir.dll[40].html, redir.dll[41].html) and Flash objects (.swf) were reconstructed, which may represent redirect attempts or exploit delivery. These findings support the possibility that sensitive credentials, personal files, and potentially malicious web content were transmitted during the observed event. However, there is not enough evidence to confirm this, as they could also be benign files.

SNORT Analysis

To confirm whether or not malicious activity took place over the network, I began to use Snort. After configuring the rules and changing the paths I ran the command:

```
"snort -c C:\Snort\etc\snort.conf -r C:\Snort\3523_Lab1_Capture_File.pcap -A console -q -K none"
```

See [Figure 3.1](#) in the Snort Analysis Appendix for a screenshot of the initial Snort console output.

Too dig deeper and confirm the safety of the network I ran:

```
"snort -c C:\Snort\etc\snort.conf -r C:\Snort\3523_Lab1_Capture_File.pcap -A fast -l C:\Snort\snort_out"
```

Snort read the PCAP and processed 2,449 packets in ~0.47s, receiving/analyzing 100% of them. See [Figure 3.2](#) in the Snort Analysis Appendix for a screenshot of Snort run time. The traffic is mixed, as it is IPv4 dominated (95%), mostly TCP (85%) with some UDP (10%) and ARP (4.9%). See [Figure 3.3](#) for a screenshot of the Snort protocol breakdown. A total of 387 alerts were triggered. However, they primarily appear to be from benign user activity. See [Figure 3.4](#) for a screenshot of Snort action stats.

No high-severity alerts (priority 1) were found. Everything appears to be priority 2 or 3 which does not require concern. The large amount of *http_inspect "server response before client request"* messages is expected when analyzing an offline PCAP that captures mid-connection or lost packets, especially during a big HTTP burst such as the one found before between 17:31 and 17:32. Additionally, I found TCP small-segment anomalies on FTP/TLS, occasional "reset outside window" events, and two portscan preprocessor portsweep alerts that are consistent with the client opening many short-lived connections to legitimate web/FTP servers. No signs of malicious network activity were observed. The capture reflects normal user activity (web browsing, HTTP downloads from 216.166.24.20, brief FTP) with no evidence of a malicious event.

Findings

a. How long did the session capture last?

- Start time: 2005-10-30 17:29:35
- End time: 2005-10-30 17:38:00
- Elapsed: 8 minutes, 25 seconds (505.697 seconds)

b. How many packets were captured?

- Packets captured: 2,449

c. How many bytes were captured?

- Total bytes: 811,157 bytes (~811 KB)

d. What protocols were observed?

- Ethernet (100%)
- IPv4 (94.9%)

- ARP (5.1%)
- UDP (10.0%)
 - DNS (3.7%)
 - NetBIOS (0.7%)
 - TiVoConnect Discovery Protocol (4.9%)
- TCP (84.9%)
 - TLS/SSL (3.6%)
 - HTTP (7.3%)
 - FTP (3.9%)
 - FTP Data (0.8%)
 - Data (8.1%)

e. When did the bulk of the data get transmitted?

- 17:31-17:32 had spikes reaching over 150 packets/sec and about 50,000-60,000 bytes/sec.
- Around 17:37:30 there was a spike reaching over 90,000 bytes/sec.

f. What caused this transmission spike?

- The first spike was caused by HTTP downloads from a server (216.166.24.20) to the local client (172.16.1.35).

g. Were any Internet Service Provider sites accessed? If so, which ones? What accounts?

- America Online (AOL) was accessed by KAUFMANUPSTAIRS.

h. What is the name of the host computer? It's IP address?

- Host Computer's Name: KAUFMANUPSTAIRS
- IP: 172.16.1.35
- MAC: 00:40:ca:70:19:a3 (First International Computer, Inc.)

i. What Operating system is it using?

- Windows XP (pc/WinNT:5.1.2600).

j. What does the local network look like?

- The LAN is 172.16.1.0/16.
- Active devices observed:
 - 172.16.1.35 — Windows XP host (KAUFMANUPSTAIRS), main client.
 - 172.16.1.1 — Default gateway/router.
 - 172.16.1.37 and 172.16.1.39 — Minimal traffic, possibly other hosts (ARP only).
- Broadcast traffic to 172.16.255.255 is present, typical of local discovery.
- The local network therefore consists of the main client, the router, and at least two additional local devices. See [Figure 1.6](#) for a screenshot of IPv4 endpoints.

k. What device names are on the local network?

- Determined from the Wireshark NBNS filter and NetworkMiner hosts list that the local device names are:
 - KAUFMANUPSTAIRS: the main Windows XP host at 172.16.1.35.
 - DVR-8525: a DVR/TiVo device at 172.16.1.37.
 - homeportal.gateway (2wire.net): the router at 172.16.0.1.
 - 172.16.1.39: Windows NT Workstation in workgroup MSHOME (no NetBIOS hostname observed).

l. Did I access any other computers on the local area network?

- Normal DHCP/DNS traffic with the router (172.16.0.1 – homeportal.gateway.2wire.net).
- Broadcast discovery with 172.16.1.37 (DVR-8525) and 172.16.1.39, but no actual session traffic between other computers on the LAN.

m. Are any other devices on the network?

- Yes. The capture shows additional LAN devices beyond the client:
 - Router: 172.16.0.1 (homeportal.gateway.2wire.net)
 - DVR/TiVo: 172.16.1.37 (DVR-8525)
 - Windows Client: 172.16.1.39 (MSHOME)

Narrative

Using the analysis, a timeline of the capture shows a single Windows XP host (KAUFMANUPSTAIRS) browsing the web for about 8 1/2 minutes on Oct 30, 2005. The host initiated an AOL install/update, fetching multiple components (some adware-like but typical for the period). The host briefly connected with a public FTP site and also conversed with common portals (Microsoft, MSN, Yahoo). The traffic bursts, TCP anomalies, and Snort alerts are best explained by legitimate downloads over HTTP rather than malicious activity.

Conclusion

Based on the packet timelines, flow directions, payload types, host roles, and IDS results, there is no explicit proof of a compromise or active attack in this capture. What looked suspicious at first is best explained by normal user activity, primarily web browsing and an AOL install/update. Potentially suspicious elements (EXE/DLL over HTTP, redirect files, anonymous FTP) are consistent with legitimate but risky practices common in 2005. In conclusion, there are potential security risks but no clear evidence of malicious activity.

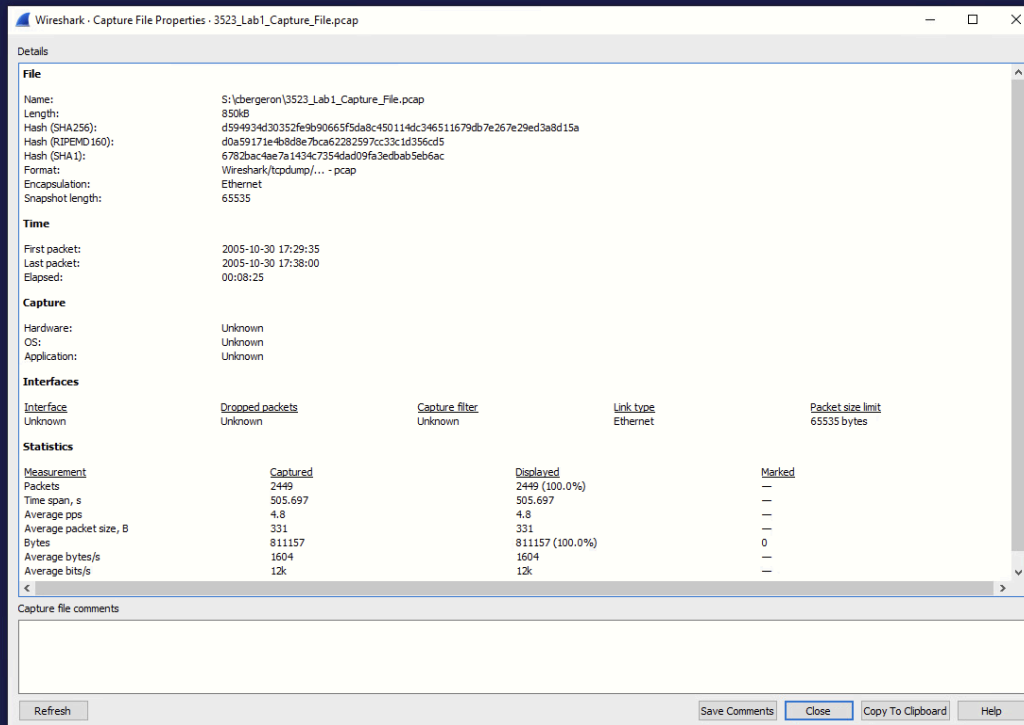
References

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- “NetworkMiner Professional Tutorial.” n.d. Netresec. <https://www.netresec.com/?page=TutorialNMP>.
- “Snort | Kali Linux Tools.” n.d. Kali Linux. <https://www.kali.org/tools/snort/>.
- “Wireshark User’s Guide.” n.d. https://www.wireshark.org/docs/wsug_html_chunked/.

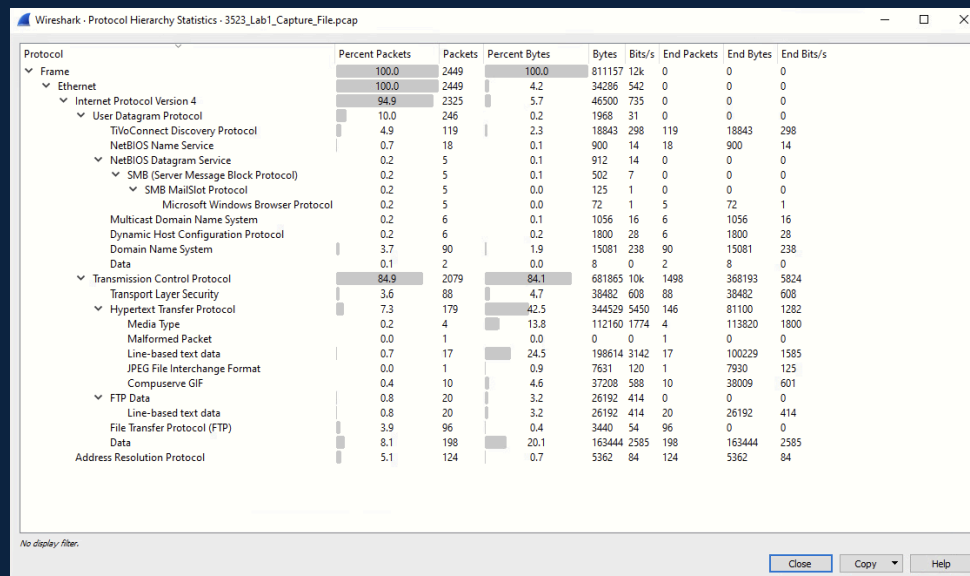
Appendix

1. Wireshark Analysis

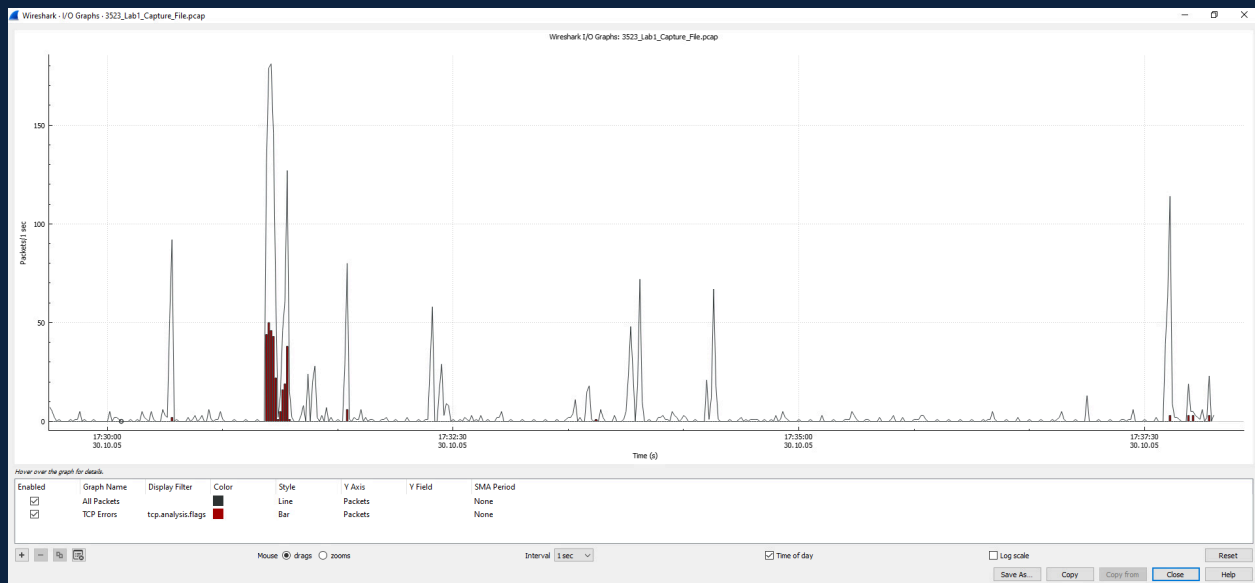
- Figure 1.1: Screenshot of Wireshark's Capture File Properties



- Figure 1.2: Screenshot of Wireshark's Protocol Hierarchy



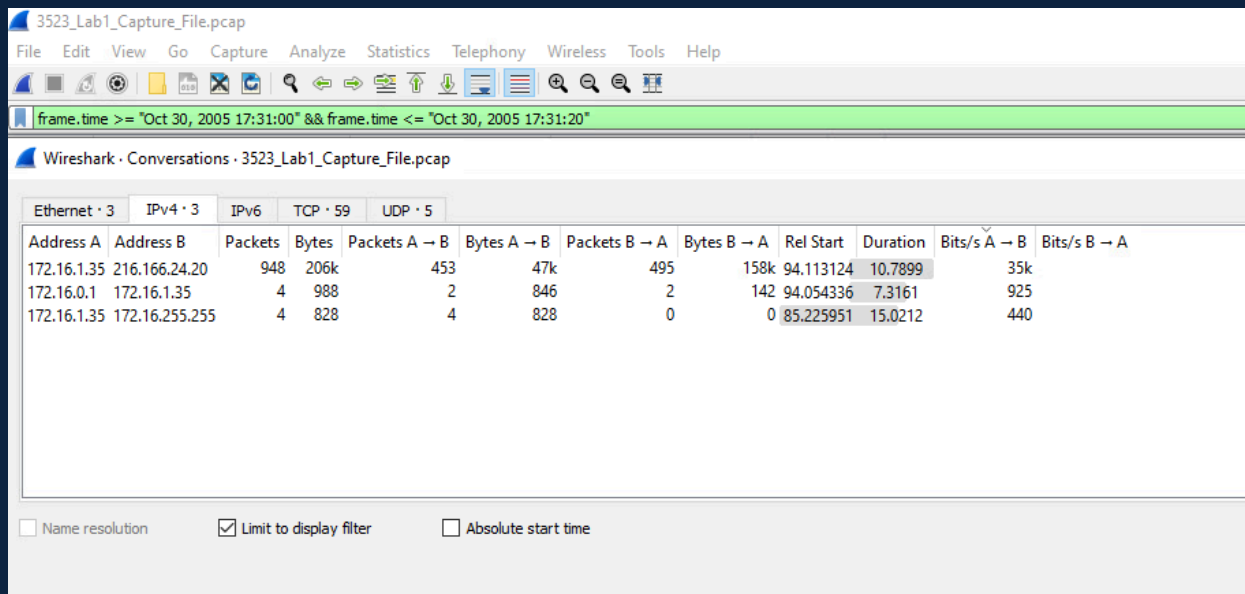
- **Figure 1.3: Screenshot of Packets Over Time**



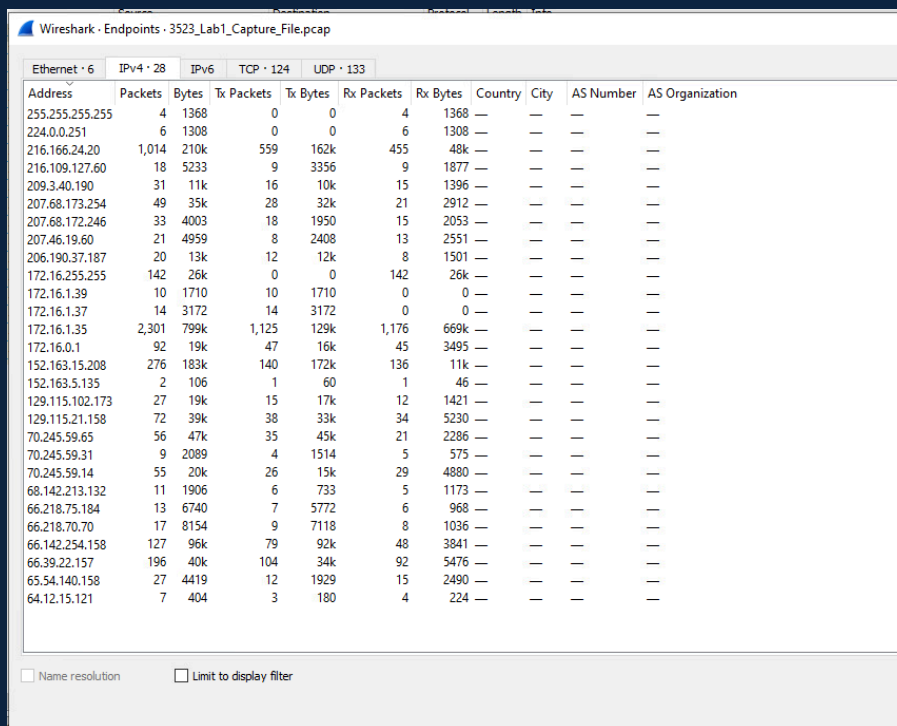
- **Figure 1.4: Screenshot of Bytes Per Second**



- **Figure 1.5: Screenshot of IPv4 Conversation During Spike**



- **Figure 1.6: Screenshot of IPv4 endpoints list**



2. NetworkMiner Analysis

- Figure 2.1: Screenshot of Credentials in NetworkMiner

Client	Server	Protocol	Username	Password	Valid login	Login timestamp
172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS-000...	66.39.22.157 [linux-wlan.org] [ftp.linux-wlan.org] (FreeBSD)	FTP	anonymous	IEUser@	Unknown	2005-10-30 21:30:27 UTC
172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS-000...	207.46.19.60 [www.microsoft.com]	HTTP Cookie	MC1=GUID=2b5cd4da54c094ab65f2a36c7b1170&HAS...	N/A	Unknown	2005-10-30 21:30:27 UTC
172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS-000...	207.68.172.246 [home.microsoft.com]	HTTP Cookie	MC1=GUID=2b5cd4da54c094ab65f2a36c7b1170&HAS...	N/A	Unknown	2005-10-30 21:30:27 UTC
172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS-000...	207.68.173.254 [www.msn.com]	HTTP Cookie	p68bd=1986177314.pf6ext=y; MC1=v+3&GUID=b39ada1...	N/A	Unknown	2005-10-30 21:31:44 UTC
172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS-000...	209.3.40.190 [fb.msn.com]	HTTP Cookie	p68bd=1986177314.pf6ext=y; MC1=v+3&GUID=b39ada1...	N/A	Unknown	2005-10-30 21:31:44 UTC
172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS-000...	65.54.140.158 [c.msn.com]	HTTP Cookie	p68bd=1986177314.pf6ext=y; MC1=v+3&GUID=b39ada1...	N/A	Unknown	2005-10-30 21:30:28 UTC
172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS-000...	206.190.37.187 [us.f812.mail.yahoo.com]	HTTP Cookie	Q=q1+AACAAAAAAbw-8q2+Q2vEag--; B=9qddk90k...	N/A	Unknown	2005-10-30 21:34:22 UTC
172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS-000...	68.142.213.132 [oc.us.yahoo.com]	HTTP Cookie	Q=q1+AACAAAAAAbw-8q2+Q2vEag--; B=9qddk90k...	N/A	Unknown	2005-10-30 21:33:51 UTC
172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS-000...	66.218.75.184 [mail.yahoo.com]	HTTP Cookie	Q=q1+AACAAAAAAbw-8q2+Q2vEag--; B=9qddk90k...	N/A	Unknown	2005-10-30 21:33:50 UTC
172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS-000...	129.115.21.158 [faculty.business.utsa.edu]	HTTP Cookie	WEBTRENDS_ID=66.142.88.176-4217656448.29657262	N/A	Unknown	2005-10-30 21:32:27 UTC
172.16.1.35 [KaufmanUpstairs] [KAUFMANUPSTAIRS-000...	206.190.37.187 [f812.mail.yahoo.com] [us.f812.mail.yahoo...	HTTP Cookie	YM.Gen=i+vaJ6l8rCyUC6SzhqzJZf6t2RF8UghzYK...	N/A	Unknown	2005-10-30 21:34:23 UTC

3. Snort Analysis

- Figure 3.1: Screenshot of Snort Console Output

```
C:\Snort\bin>snort -c C:\Snort\etc\snort.conf -r C:\Snort\3523_Lab1_Capture_File.pcap -A console -q -k none
10/30/17:30:27.357070 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 207.46.19.60:80 -> 172.16.1.35:3371
10/30/17:30:27.428392 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 207.68.172.246:80 -> 172.16.1.35:3372
10/30/17:30:27.595580 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 207.68.173.254:80 -> 172.16.1.35:3373
10/30/17:30:27.877602 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 207.68.173.254:80 -> 172.16.1.35:3373
10/30/17:30:27.885441 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 207.68.173.254:80 -> 172.16.1.35:3373
10/30/17:30:27.989937 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 207.68.173.254:80 -> 172.16.1.35:3373
10/30/17:30:27.999781 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 207.68.173.254:80 -> 172.16.1.35:3373
10/30/17:30:28.007699 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 207.68.173.254:80 -> 172.16.1.35:3373
10/30/17:30:28.017529 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 207.68.173.254:80 -> 172.16.1.35:3373
10/30/17:30:28.104263 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 207.68.173.254:80 -> 172.16.1.35:3373
10/30/17:30:28.168809 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 207.68.173.254:80 -> 172.16.1.35:3373
10/30/17:30:28.358624 ** [129:12:2] (conntrack) Consecutive TCP small segments exceeding threshold ** [Classification: Potentially Bad Traffic] [Priority: 2] (TCP) 66.39.22.157:21 -> 172.16.1.35:3370
10/30/17:30:28.445220 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 65.54.140.158:80 -> 172.16.1.35:3374
10/30/17:30:28.520026 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.524101 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.557709 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.567087 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.575413 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.604978 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.612088 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.619078 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.656229 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.662103 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.671987 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.679073 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.687796 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.697638 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.705234 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.713426 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.721278 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.731143 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.739112 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.747216 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 66.142.254.158:80 -> 172.16.1.35:3377
10/30/17:30:28.750754 ** [120:15:2] (reset_outside) Reset outside window ** [Classification: Potentially Bad Traffic] [Priority: 2] (TCP) 207.46.19.60:80 -> 172.16.1.35:3372
10/30/17:30:28.752764 ** [120:15:2] (reset_outside) Reset outside window ** [Classification: Potentially Bad Traffic] [Priority: 2] (TCP) 66.39.22.157:61000 -> 172.16.1.35:3376
10/30/17:30:44.157044 ** [120:15:2] (reset_outside) Reset outside window ** [Classification: Potentially Bad Traffic] [Priority: 2] (TCP) 65.54.140.158:80 -> 172.16.1.35:3374
10/30/17:30:44.157064 ** [122:1:1] (portscan) TCP Portscan ** [Classification: Attempted Information Leak] [Priority: 2] (PROTO:255) 172.16.1.35 -> 65.54.140.158
10/30/17:30:44.157092 ** [120:15:2] (reset_outside) Reset outside window ** [Classification: Potentially Bad Traffic] [Priority: 2] (TCP) 66.39.22.157:21 -> 172.16.1.35:3370
10/30/17:31:09.296742 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 216.166.24.20:80 -> 172.16.1.35:3408
10/30/17:31:09.531326 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 216.166.24.20:80 -> 172.16.1.35:3409
10/30/17:31:09.538597 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 216.166.24.20:80 -> 172.16.1.35:3410
10/30/17:31:09.548785 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 216.166.24.20:80 -> 172.16.1.35:3411
10/30/17:31:09.764238 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 216.166.24.20:80 -> 172.16.1.35:3412
10/30/17:31:09.928456 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 216.166.24.20:80 -> 172.16.1.35:3413
10/30/17:31:10.008289 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 216.166.24.20:80 -> 172.16.1.35:3414
10/30/17:31:10.024118 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 216.166.24.20:80 -> 172.16.1.35:3415
10/30/17:31:10.183867 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 216.166.24.20:80 -> 172.16.1.35:3416
10/30/17:31:10.193694 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 216.166.24.20:80 -> 172.16.1.35:3416
10/30/17:31:10.201508 ** [120:18:3] (http_inspect) PROTOCOL-OTHER HTTP server response before client request ** [Classification: Unknown Traffic] [Priority: 3] (TCP) 216.166.24.20:80 -> 172.16.1.35:3416
```

- Figure 3.2: Screenshot of Snort Run Time

```
=====  
Run time for packet processing was 0.47000 seconds  
Snort processed 2449 packets.  
Snort ran for 0 days 0 hours 0 minutes 0 seconds  
Pkts/sec: 2449  
=====  
Packet I/O Totals:  
Received: 2449  
Analyzed: 2449 (100.000%)  
Dropped: 0 (0.000%)  
Filtered: 0 (0.000%)  
Outstanding: 0 (0.000%)  
Injected: 0  
=====
```

- **Figure 3.3: Screenshot of Snort Protocol Breakdown**

```
=====
Breakdown by protocol (includes rebuilt packets):
  Eth:      2534 (100.000%)
  VLAN:     0 ( 0.000%)
  IP4:      2410 ( 95.107%)
  Frag:     0 ( 0.000%)
  ICMP:     0 ( 0.000%)
  UDP:      246 (  9.708%)
  TCP:     2164 ( 85.399%)
  IP6:      0 ( 0.000%)
  IP6 Ext:  0 ( 0.000%)
  IP6 Opts: 0 ( 0.000%)
  Frag6:    0 ( 0.000%)
  ICMP6:    0 ( 0.000%)
  UDP6:     0 ( 0.000%)
  TCP6:     0 ( 0.000%)
  Teredo:   0 ( 0.000%)
  ICMP-IP:  0 ( 0.000%)
  EAPOL:    0 ( 0.000%)
  IP4/IP4:  0 ( 0.000%)
  IP4/IP6:  0 ( 0.000%)
  IP6/IP4:  0 ( 0.000%)
  IP6/IP6:  0 ( 0.000%)
  GRE:      0 ( 0.000%)
  GRE Eth:  0 ( 0.000%)
  GRE VLAN: 0 ( 0.000%)
  GRE IP4:  0 ( 0.000%)
  GRE IP6:  0 ( 0.000%)
  GRE IP6 Ext: 0 ( 0.000%)
  GRE PPTP: 0 ( 0.000%)
  GRE ARP:  0 ( 0.000%)
  GRE IPX:  0 ( 0.000%)
  GRE Loop: 0 ( 0.000%)
  MPLS:     0 ( 0.000%)
  ARP:      124 (  4.893%)
  IPX:      0 ( 0.000%)
  Eth Loop: 0 ( 0.000%)
  Eth Disc: 0 ( 0.000%)
  IP4 Disc: 0 ( 0.000%)
  IP6 Disc: 0 ( 0.000%)
  TCP Disc: 0 ( 0.000%)
  UDP Disc: 0 ( 0.000%)
  ICMP Disc: 0 ( 0.000%)
  All Discard: 0 ( 0.000%)
  Other:    0 ( 0.000%)
  Bad Chk Sum: 821 (32.399%)
  Bad TTL:  0 ( 0.000%)
  S5 G 1:   85 (  3.354%)
  S5 G 2:   0 ( 0.000%)
  Total:    2534
=====
```

- **Figure 3.4: Screenshot of Snort Action Stats**

```
=====
Action Stats:
  Alerts:      387 ( 15.272%)
  Logged:     1082 ( 42.699%)
  Passed:      0 (  0.000%)
Limits:
  Match:      0
  Queue:      0
  Log:        0
  Event:      0
  Alert:      0
Verdicts:
  Allow:      2392 ( 97.673%)
  Block:      0 (  0.000%)
  Replace:    0 (  0.000%)
  Whitelist:  57 (  2.327%)
  Blacklist:  0 (  0.000%)
  Ignore:     0 (  0.000%)
  (null):     0 (  0.000%)
=====
```